```
import pandas as pd
import random
import matplotlib.pyplot as plt

study_data = {
    "Topic": ["Artificial Intelligence", "Python", "Machine Learning"],
    "Content": [
        "AI is the simulation of human intelligence in machines.",
        "Python is a powerful programming language used in AI and data science.",
        "Machine Learning is a subset of AI that allows systems to learn from data."
    ]
}

study_df = pd.DataFrame(study_data)
print("Study database loaded successfully!")
```

```
Study database loaded successfully!
```

```
quiz_data = {
    "question": [
        "What does AI stand for?",
        "Python is a ___?",
        "Machine Learning is part of ___?"
    ],

    "option1": [
        "Artificial Intelligence",
        "Programming Language",
        "AI"
    ],

    "option2": [
        "Automatic Input",
        "Animal",
        "Math"
    ],

    "option3": [
        "Internet Access",
        "Browser",
        "Physics"
    ],

 "option4": [
        "None",
        "Game",
        "Chemistry"
    ],

    "answer": [
        "Artificial Intelligence",
        "Programming Language",
        "AI"
    ]
}

quiz_df = pd.DataFrame(quiz_data)

print(quiz_df)
```

```
                           question                  option1          option2  \
0           What does AI stand for?  Artificial Intelligence  Automatic Input
1                  Python is a ___?     Programming Language           Animal
2  Machine Learning is part of ___?                       AI             Math

           option3    option4                   answer
0  Internet Access       None  Artificial Intelligence
1          Browser       Game     Programming Language
2          Physics  Chemistry                       AI
```

```
score = 0

for i in range(len(quiz_df)):

    print("\nQuestion:", quiz_df.loc[i, "question"])

    print("1.", quiz_df.loc[i, "option1"])
    print("2.", quiz_df.loc[i, "option2"])
    print("3.", quiz_df.loc[i, "option3"])
```

```
    print("4.", quiz_df.loc[i, "option4"])

    answer = input("Enter option number (1-4): ")

    options = {
        "1": quiz_df.loc[i, "option1"],
        "2": quiz_df.loc[i, "option2"],
        "3": quiz_df.loc[i, "option3"],
        "4": quiz_df.loc[i, "option4"]
    }

    if options.get(answer, "").lower() == quiz_df.loc[i, "answer"].lower():
        print("Correct!")
        score += 1
    else:
        print("Wrong! Correct answer:", quiz_df.loc[i, "answer"])

print("\nFinal Score:", score)
```

```
Question: What does AI stand for?
1. Artificial Intelligence
2. Automatic Input
3. Internet Access
4. None
Enter option number (1-4):  1
Correct!

Question: Python is a ___?
1. Programming Language
2. Animal
3. Browser
4. Game
Enter option number (1-4):  1
Correct!

Question: Machine Learning is part of ___?
1. AI
2. Math
3. Physics
4. Chemistry
Enter option number (1-4):  1
Correct!

Final Score: 3
```

```
def study_topic():

    topic = input("Enter topic to study: ")

    results = study_df[study_df["topic"].str.contains(topic, case=False)]

    if not results.empty:
        print("\nStudy Material:\n")
        print(results["content"].values[0])
    else:
        print("Topic not found.")
```

```
def study_topic():

    topic = input("Enter topic to study: ")

    results = study_df[study_df["topic"].str.contains(topic, case=False)]

    if not results.empty:
        print("\nStudy Material:\n")
        print(results["content"].values[0])
    else:
        print("Topic not found.")
```

```
def start_quiz():
    difficulty = input("Choose difficulty (Easy/Medium/Hard): ")
    questions = quiz_df[quiz_df["difficulty"].str.lower() == difficulty.lower()]
    questions = questions.sample(frac=1).reset_index(drop=True)  # Fixed indentation here

    score = 0

    for i in range(len(questions)):  # Changed quiz_df to questions to match the filtered dataset

        print("\nQuestion:", questions.loc[i,"question"])  # Changed quiz_df to questions

        print("1.", questions.loc[i,"option1"])  # Changed quiz_df to questions
```

```
        print("2.", questions.loc[i,"option2"])  # Changed quiz_df to questions
        print("3.", questions.loc[i,"option3"])  # Changed quiz_df to questions
        print("4.", questions.loc[i,"option4"])  # Changed quiz_df to questions

        answer = input("Enter your answer: ")

        if answer.lower() == questions.loc[i,"answer"].lower():  # Changed quiz_df to questions
            print("Correct!")
            score += 1
        else:
            print("Wrong!")

    show_feedback(score, len(questions))
    score_history.append(score)
```

```
# First, make sure all the functions are defined before the menu loop
def study_topic():
    # Initialize a dictionary with the 'topic' key if it doesn't exist
    topics = {"topic": "default_topic"}  # Add default value or prompt user for input
    # Rest of the function code...
    print(f"Studying topic: {topics['topic']}")

def start_quiz():
    # Your quiz function code...
    pass

def show_stats():
    # Your stats function code...
    pass

def recommend_topic():
    # Make sure this function also has proper initialization of any dictionaries
    topics = {"topic": "recommended_topic"}  # Add default value
    # Rest of the function code...
    print(f"Recommended topic: {topics['topic']}")

def show_graph():
    # Your graph function code...
    pass

def generate_question():
    # Your question generation code...
    pass

def adaptive_learning():
    # Your adaptive learning code...
    pass

def smart_recommendation():
    # Your smart recommendation code...
    pass

def study_chatbot():
    # Your chatbot code...
    pass

def leaderboard():
    # Your leaderboard code...
    pass

# Now the menu loop
while True:
    print("\n--- AI Study Pal ---")
    print("1. Study Topic")
    print("2. Take Quiz")
    print("3. View Progress")
    print("4. Get Recommendation")
    print("5. Show performance graph")
    print("6. Generate AI question")
    print("7. Adaptive learning report")
    print("8. Exit")
    print("9. Smart Recommendation")  # Added missing menu options
    print("10. Study Chatbot")
    print("11. Leaderboard")

    choice = input("Choose an option: ")

    if choice == "1":
        study_topic()

    elif choice == "2":
        start_quiz()
```

```
    elif choice == "3":
        show_stats()

    elif choice == "4":
        recommend_topic()

    elif choice == "5":
        show_graph()

    elif choice == "6":
        generate_question()

    elif choice == "7":
        adaptive_learning()

    elif choice == "8":
        break  # Exit the program

    elif choice == "9":
        smart_recommendation()

    elif choice == "10":
        study_chatbot()

    elif choice == "11":
        leaderboard()

    else:
        print("Invalid choice")
```

```
--- AI Study Pal ---
1. Study Topic
2. Take Quiz
3. View Progress
4. Get Recommendation
5. Show performance graph
6. Generate AI question
7. Adaptive learning report
8. Exit
9. Smart Recommendation
10. Study Chatbot
11. Leaderboard
Choose an option:  1
Studying topic: default_topic

--- AI Study Pal ---
1. Study Topic
2. Take Quiz
3. View Progress
4. Get Recommendation
5. Show performance graph
6. Generate AI question
7. Adaptive learning report
8. Exit
9. Smart Recommendation
10. Study Chatbot
11. Leaderboard
Choose an option:  2

--- AI Study Pal ---
1. Study Topic
2. Take Quiz
3. View Progress
4. Get Recommendation
5. Show performance graph
6. Generate AI question
7. Adaptive learning report
8. Exit
9. Smart Recommendation
10. Study Chatbot
11. Leaderboard
Choose an option:  3

--- AI Study Pal ---
1. Study Topic
2. Take Quiz
3. View Progress
4. Get Recommendation
5. Show performance graph
6. Generate AI question
7. Adaptive learning report
8. Exit
9. Smart Recommendation
```

```
10. Study Chatbot
11. Leaderboard
Choose an option:  4
Recommended topic: recommended topic
```

```
import random
```

```
import random
```

```
def start_quiz():

    score = 0

    questions = quiz_df.sample(frac=1).reset_index(drop=True)

    for i in range(len(questions)):

        print("\nQuestion:", questions.loc[i,"question"])

        print("1.", questions.loc[i,"option1"])
        print("2.", questions.loc[i,"option2"])
        print("3.", questions.loc[i,"option3"])
        print("4.", questions.loc[i,"option4"])

        answer = input("Enter your answer: ")

        if answer.lower() == questions.loc[i,"answer"].lower():
            print("Correct!")
            score += 1
        else:
            print("Wrong!")

    print("\nFinal Score:", score)
    save_progress(score)
```

```
def show_feedback(score, total):
    percentage = (score/total) * 100

    print("\nYour Score:", score, "/", total)

    if percentage >= 80:
        print("Excellent! 🌟")
    elif percentage >= 50:
        print("Good job 👍")
    else:
        print("Keep practicing 📚")
        # Note: This line was using 'questions' which isn't defined in this snippet
        # percentage = (score/len(questions))*100

    # Moving these checks inside the function so they can access 'percentage'
    if percentage > 80:
        print("You are doing great! Try a harder level next time.")
    elif percentage < 40:
        print("You should practice easier questions first.")
```

```
import json
```

```
def save_progress(score):

    try:
        with open("progress.json","r") as file:
            data = json.load(file)
    except:
        data = {}

    if username not in data:
        data[username] = {"quizzes_taken":0,"best_score":0}

    data[username]["quizzes_taken"] += 1

    if score > data[username]["best_score"]:
        data[username]["best_score"] = score

    with open("progress.json","w") as file:
        json.dump(data,file)

    print("Progress saved for", username)
```

```
def show_stats():

    try:
        with open("progress.json","r") as file:
            data = json.load(file)

        if username in data:

            print("\nUser:", username)
            print("Quizzes Taken:", data[username]["quizzes_taken"])
            print("Best Score:", data[username]["best_score"])

        else:
            print("No data for this user yet.")

    except:
        print("No statistics available.")
```

```
# Import pandas library first
import pandas as pd

quiz_data = {
    "topic": ["Python", "Python", "AI"],
    "question": ["What is a list?", "What is a dictionary?", "What is AI?"],
    "option1": ["Collection", "Key-value pair", "Artificial Intelligence"],
    "option2": ["Number", "Loop", "Animal Intelligence"],
    "option3": ["Boolean", "Variable", "Auto Input"],
    "option4": ["String", "Condition", "None"],
    "answer": ["Collection", "Key-value pair", "Artificial Intelligence"]
}

# Now pd is defined and can be used
quiz_df = pd.DataFrame(quiz_data)
```

```
topic_scores = {}
```

```
def start_quiz():

    score = 0
    questions = quiz_df.sample(frac=1).reset_index(drop=True)

    for i in range(len(questions)):

        topic = questions.loc[i, "topic"]

        print("\nTopic:", topic)
        print("Question:", questions.loc[i, "question"])

        print("1.", questions.loc[i,"option1"])
        print("2.", questions.loc[i,"option2"])
        print("3.", questions.loc[i,"option3"])
        print("4.", questions.loc[i,"option4"])

        answer = input("Enter your answer: ")

        if topic not in topic_scores:
            topic_scores[topic] = {"correct":0,"total":0}

        topic_scores[topic]["total"] += 1

        if answer.lower() == questions.loc[i,"answer"].lower():
            print("Correct!")
            score += 1
            topic_scores[topic]["correct"] += 1
        else:
            print("Wrong!")

    print("\nFinal Score:", score)
    recommend_topic(user_score)
```

```
def recommend_topic():

    if not topic_scores:
        print("No quiz data yet.")
        return

    weakest_topic = None
    lowest_accuracy = 100
```

```
    for topic, stats in topic_scores.items():

        accuracy = (stats["correct"] / stats["total"]) * 100

        if accuracy < lowest_accuracy:
            lowest_accuracy = accuracy
            weakest_topic = topic

    print("\nRecommended Topic to Study:", weakest_topic)
    print("Accuracy:", round(lowest_accuracy,2), "%")
```

```
import matplotlib.pyplot as plt
```

```
score_history = []
```

```
def show_graph():

    if len(score_history) == 0:
        print("No quiz data yet.")
        return

    plt.plot(score_history, marker='o')

    plt.title("Quiz Score Progress")
    plt.xlabel("Quiz Attempt")
    plt.ylabel("Score")

    plt.show()
```

```
quiz_data = {
    "topic": ["Python","Python","AI","AI","Python"],
    "difficulty": ["Easy","Medium","Easy","Hard","Hard"],
    "question": [
        "What is a list?",
        "What is a dictionary?",
        "What does AI stand for?",
        "What is machine learning?",
        "What is a lambda function?"
    ],
    "option1": [
        "Collection",
        "Key-value pair",
        "Artificial Intelligence",
        "A subset of AI",
        "Anonymous function"
    ],
    "option2": [
        "Number",
        "Loop",
        "Animal Intelligence",
        "Robot learning",
        "Loop"
    ],
    "option3": [
        "Boolean",
        "Variable",
        "Auto Input",
        "Database",
        "Class"
    ],
    "option4": [
        "String",
        "Condition",
        "Artificial Input",
        "Sorting",
        "Object"
    ],
    "answer": [
        "Collection",
        "Key-value pair",
        "Artificial Intelligence",
        "A subset of AI",
        "Anonymous function"
    ]
}

quiz_df = pd.DataFrame(quiz_data)
```

```
import random

def generate_question():

    topic = input("Enter topic to generate question from: ")

    results = study_df[study_df["topic"].str.contains(topic, case=False)]

    if results.empty:
        print("Topic not found.")
        return

    text = results["content"].values[0]

    words = text.split()

    if len(words) < 6:
        print("Not enough content to generate question.")
        return

    keyword = random.choice(words)

    question = f"What is related to '{keyword}' in {topic}?"

    print("\nGenerated Question:")
    print(question)

    user_answer = input("Your answer: ")

    print("Sample reference from study material:")
    print(text[:120] + "...")
```

```
import time

def loading():
    print("Loading AI Study Pal...")
    for i in range(3):
        print(".")
        time.sleep(1)
```

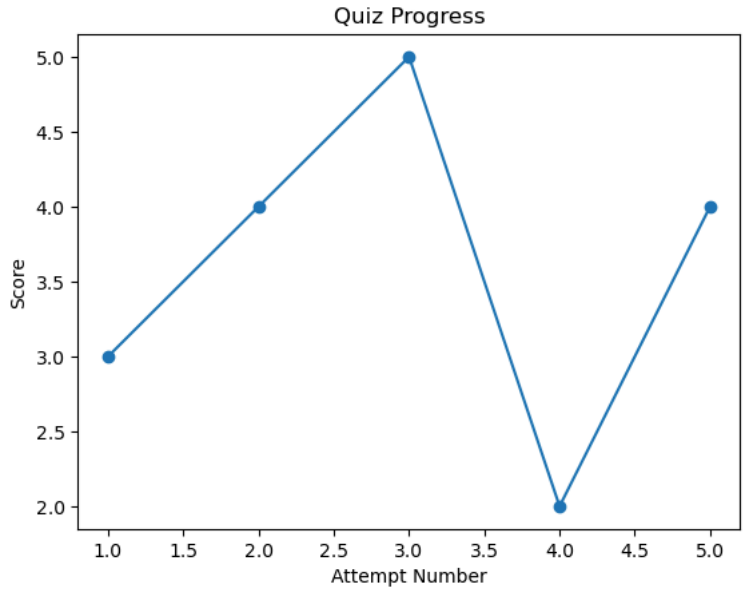
```
def leaderboard():

    try:
        with open("progress.json","r") as file:
            data = json.load(file)

        print("\nLeaderboard\n")

        for user,stats in data.items():
            print(user,"Best Score:",stats["best_score"])

    except:
        print("No leaderboard data yet.")
```

```
def smart_recommendation():

    print("\nAI Study Recommendation\n")

    if not topic_scores:
        print("No quiz data available. Take a quiz first.")
        return

    weak_topics = []

    for topic, stats in topic_scores.items():

        accuracy = (stats["correct"] / stats["total"]) * 100

        if accuracy < 50:
            weak_topics.append(topic)

    if weak_topics:
        print("You should focus on these topics:")
        for topic in weak_topics:
            print("-", topic)

        print("\nSuggested Study Plan:")
        print("1. Review study material")
        print("2. Take EASY quiz")
        print("3. Attempt MEDIUM quiz")
```

```
    else:
        print("Great job! You are performing well in all topics.")
        print("Try HARD quizzes to master the subjects.")
```

```
def study_chatbot():

    question = input("Ask a study question: ")

    results = study_df[study_df["content"].str.contains(question, case=False)]

    if results.empty:
        print("Sorry, I could not find the answer in study material.")
    else:
        print("\nAI Study Assistant Answer:\n")
        print(results["content"].values[0][:200], "...")
```

```
import matplotlib.pyplot as plt

scores = [3, 4, 5, 2, 4]   # Example quiz scores
attempts = [1, 2, 3, 4, 5]

plt.plot(attempts, scores, marker='o')
plt.title("Quiz Progress")
plt.xlabel("Attempt Number")
plt.ylabel("Score")
plt.show()
```




```
def recommend_topic(score):
    if score <= 2:
        print("Recommended: Review Python basics.")
    elif score <= 4:
        print("Recommended: Study Machine Learning.")
    else:
        print("Recommended: Move to Advanced AI.")
```

```
import random

questions = [
    "What is Artificial Intelligence?",
    "What is Python mainly used for?",
    "What is Machine Learning?",
    "Give one application of AI.",
    "What is data science?"
]

def generate_ai_question():
    q = random.choice(questions)
    print("AI Generated Question:")
    print(q)

generate_ai_question()
```

```
AI Generated Question:
What is data science?
```

```
import tkinter as tk
from tkinter import messagebox

def study_topic():
    messagebox.showinfo("Study Topic", "Topics:\n1. Artificial Intelligence\n2. Python\n3. Machine Learning")

def take_quiz():
    messagebox.showinfo("Quiz", "Quiz feature running!")

def view_progress():
    messagebox.showinfo("Progress", "Progress tracking feature!")

def ai_question():
    messagebox.showinfo("AI Question", "What is Artificial Intelligence?")

# Create window
app = tk.Tk()
app.title("AI Study Pal")
app.geometry("400x300")

title = tk.Label(app, text="AI Study Pal", font=("Arial", 18))
title.pack(pady=20)

btn1 = tk.Button(app, text="Study Topic", command=study_topic, width=20)
btn1.pack(pady=5)

btn2 = tk.Button(app, text="Take Quiz", command=take_quiz, width=20)
btn2.pack(pady=5)

btn3 = tk.Button(app, text="View Progress", command=view_progress, width=20)
btn3.pack(pady=5)

btn4 = tk.Button(app, text="AI Question Generator", command=ai_question, width=20)
btn4.pack(pady=5)

app.mainloop()
```

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.